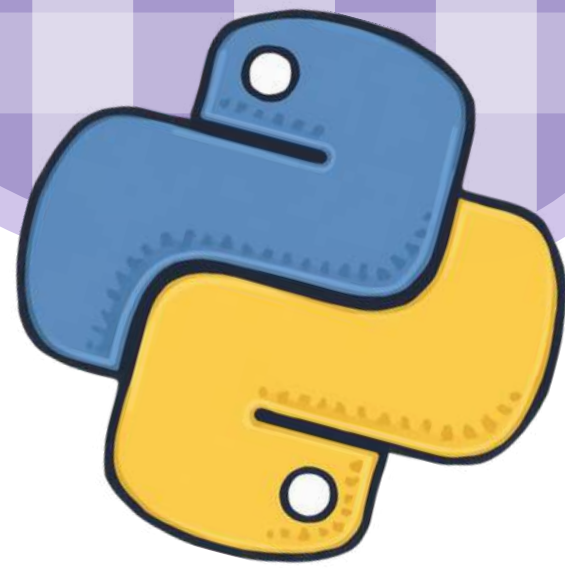
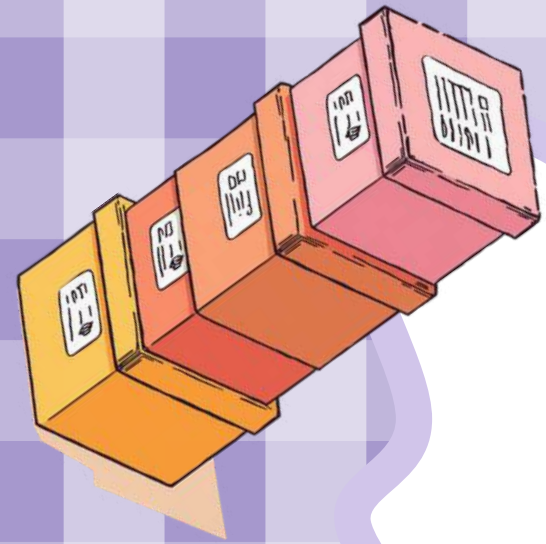
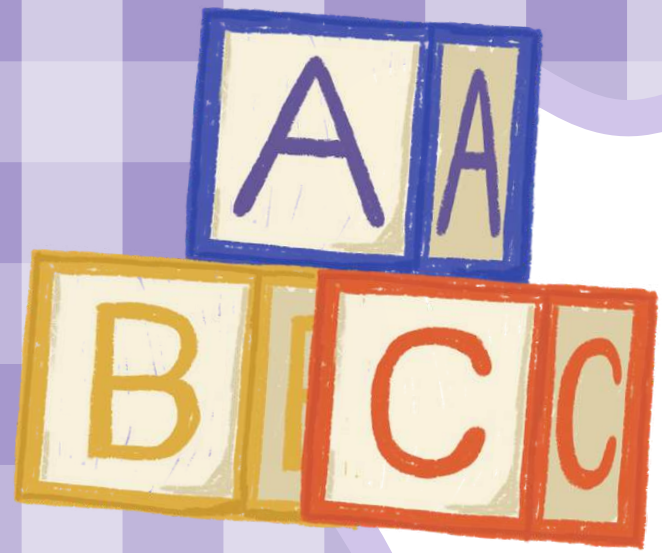




TUPLES & SETS

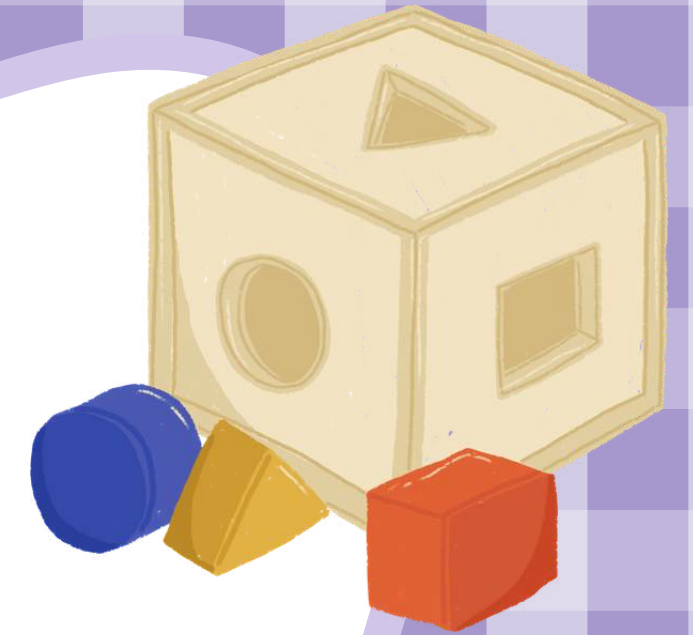
PYTHON'S SPECIAL COLLECTIONS

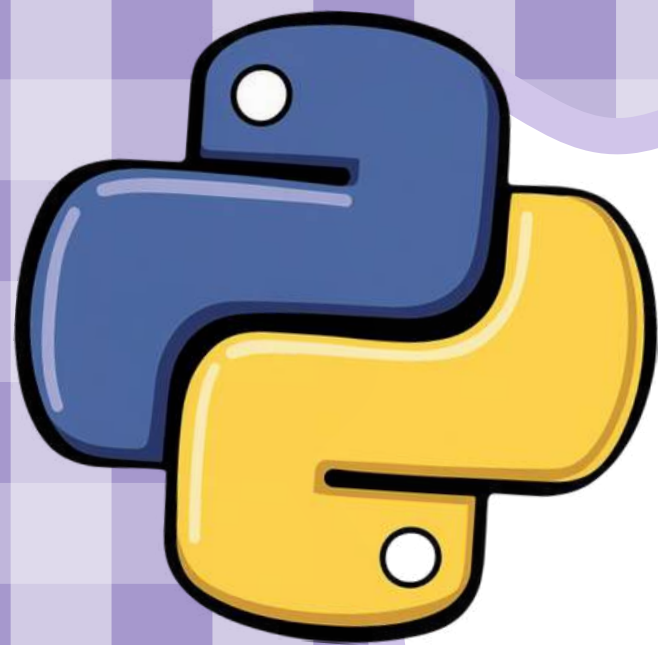




WHAT ARE COLLECTIONS IN PYTHON?

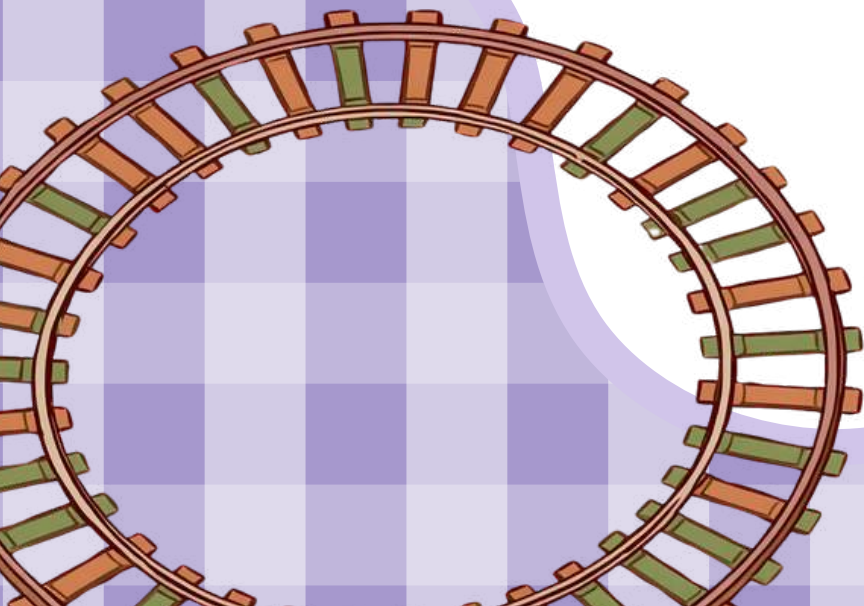
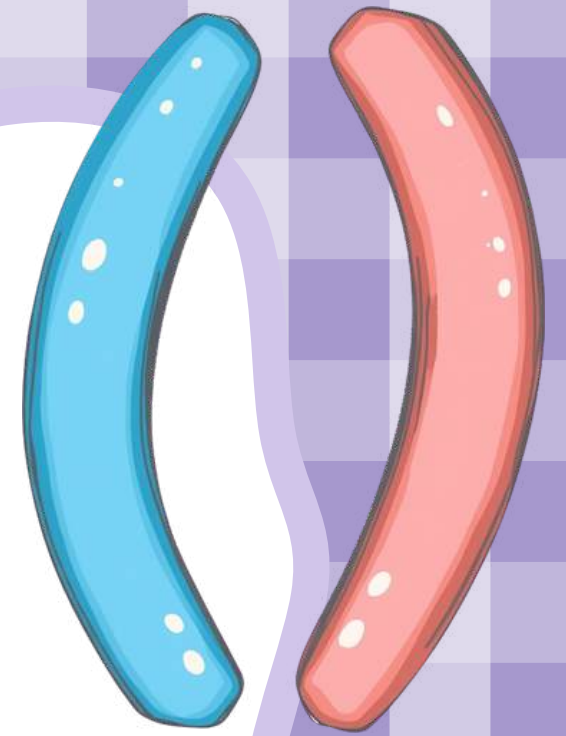
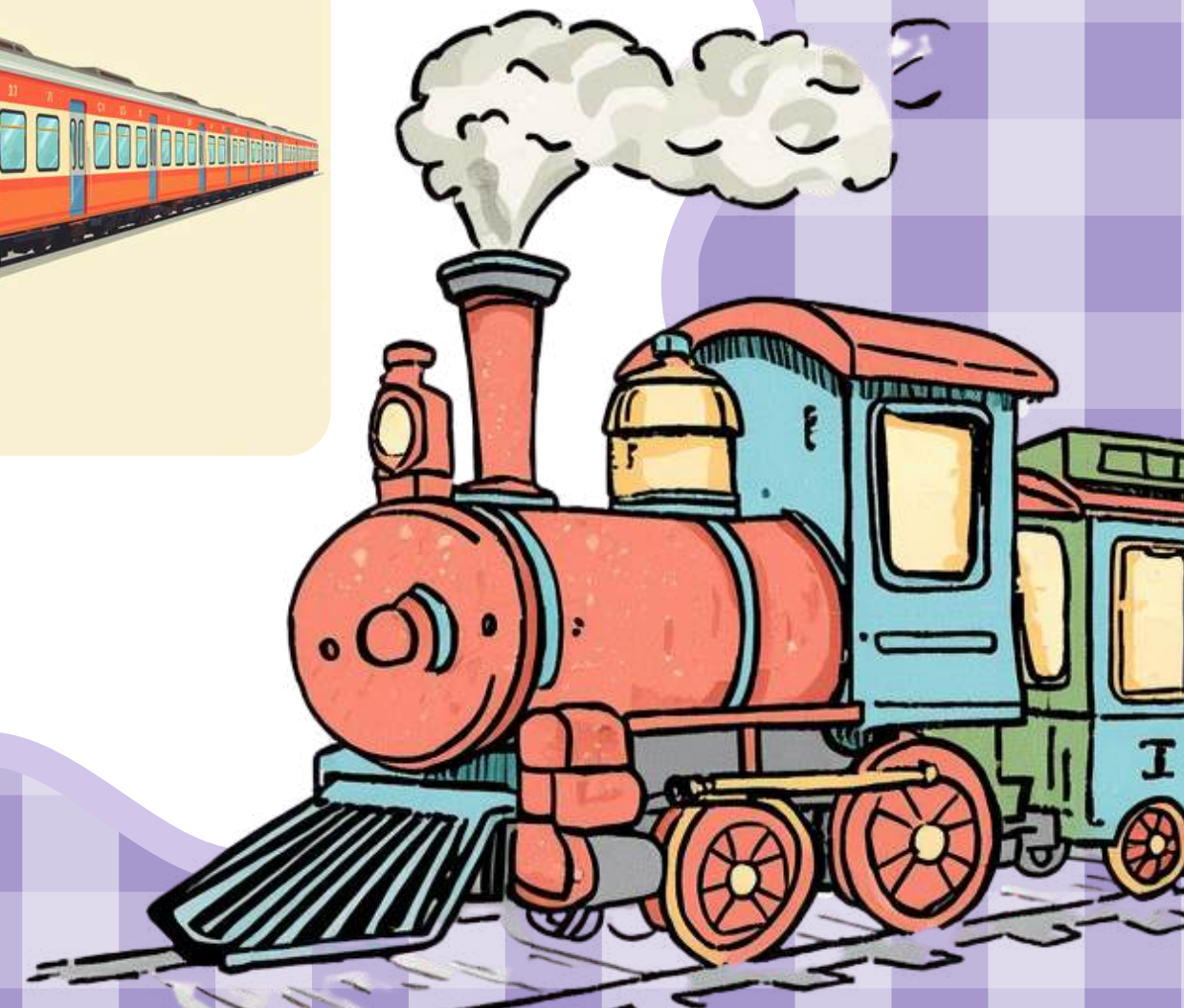
Collections are groups of items stored together in Python! Think of it like organizing your things — you can list your toys, your school books, or your friends' names all in one place.

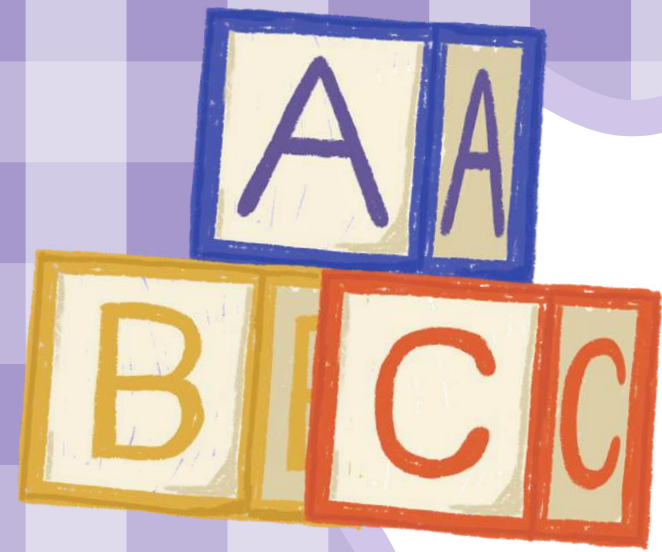




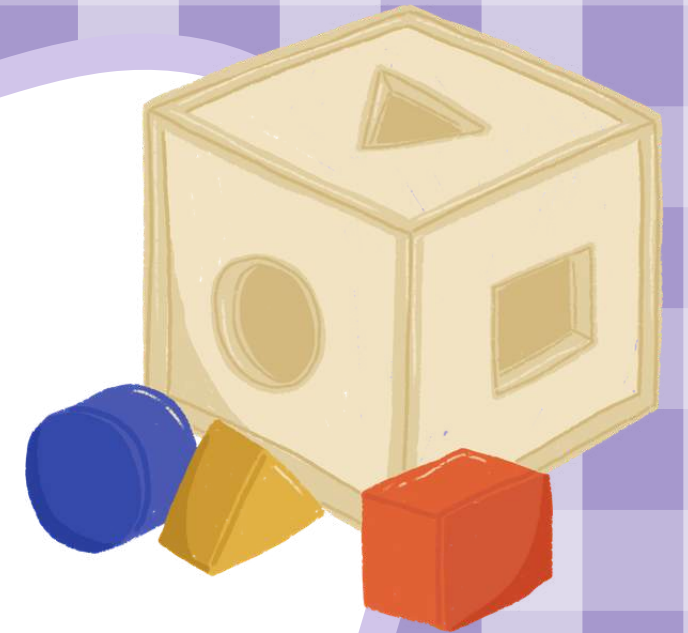
WHAT IS A TUPLE?

A tuple is a collection of items that stays in order and cannot be changed. Think of it like a train with fixed coaches — once the coaches are set, they stay in the same order!





TUPLE ANALOGY — THE TRAIN COMPARTMENT

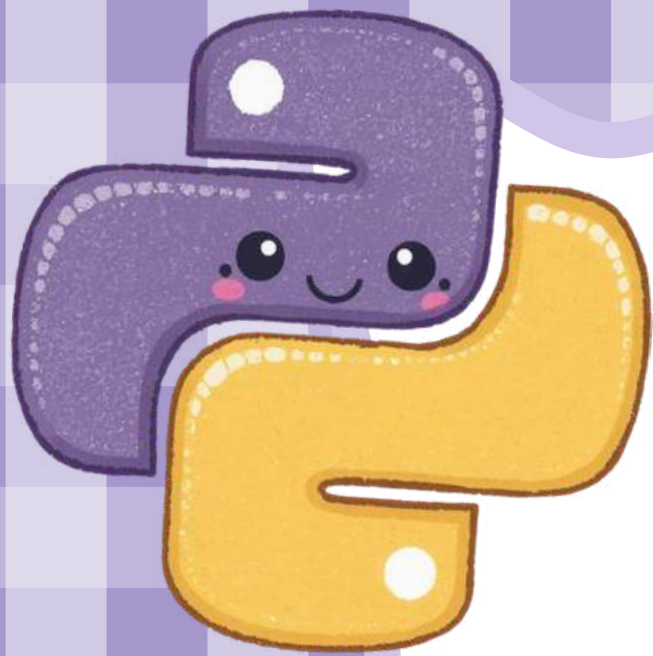


Think of the Rajdhani Express train! The coaches S1, S2, S3, AC1, AC2 are always in the same order. You cannot swap or rearrange them — just like a tuple!

Once the train is set, the order stays the same forever.

Another example: Your Aadhaar card has your name, date of birth, and number that never change — just like tuple data!



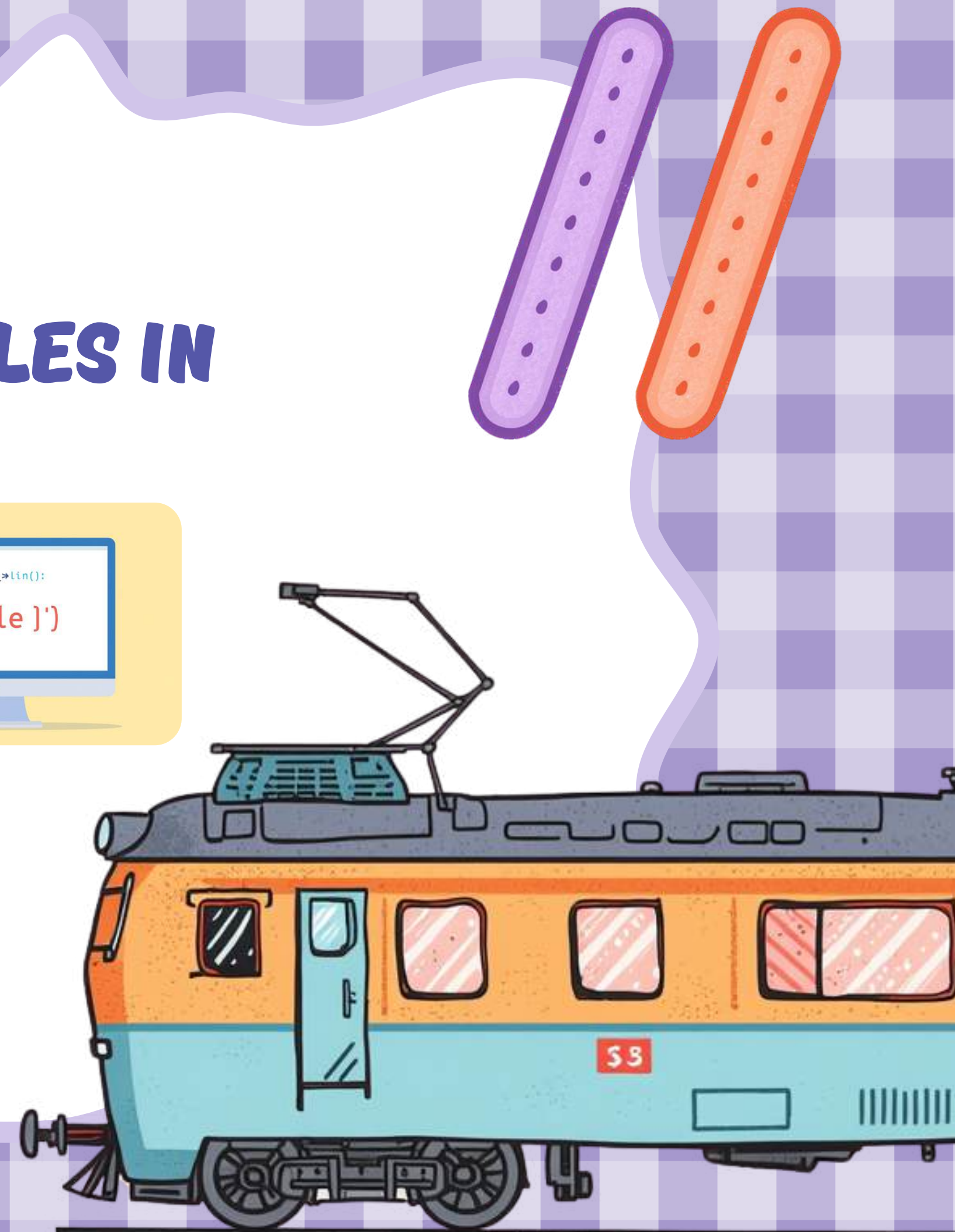
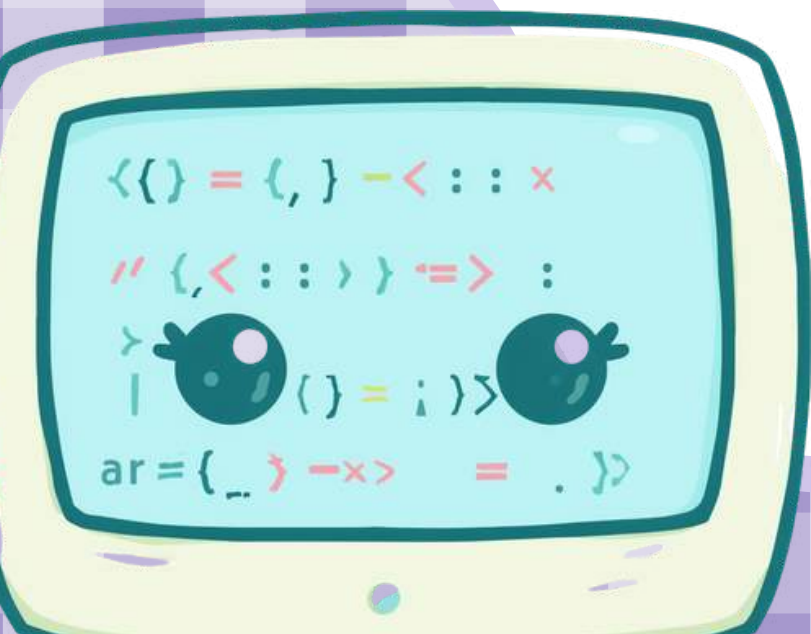
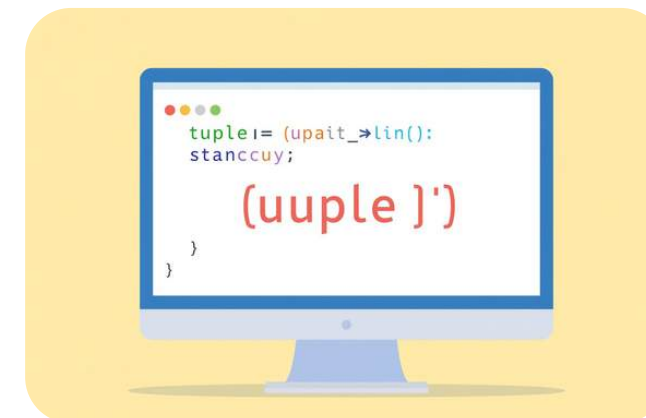


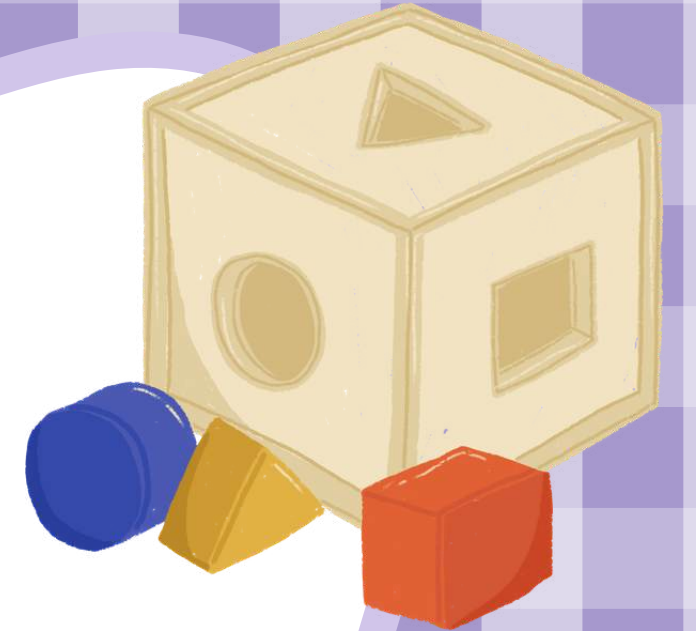
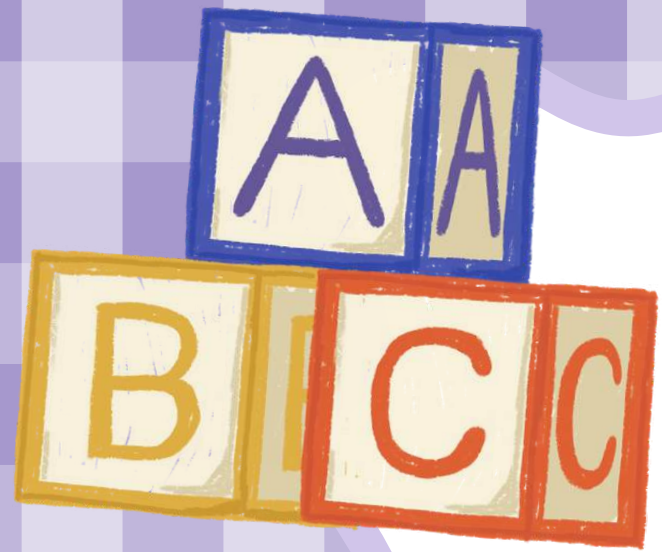
CREATING TUPLES IN PYTHON

Here's how to create a tuple:

```
train_coaches = ("S1",  
"S2", "S3", "AC1", "AC2")  
print(train_coaches)
```

- Use parentheses ()
- Items separated by commas
- Order stays the same!

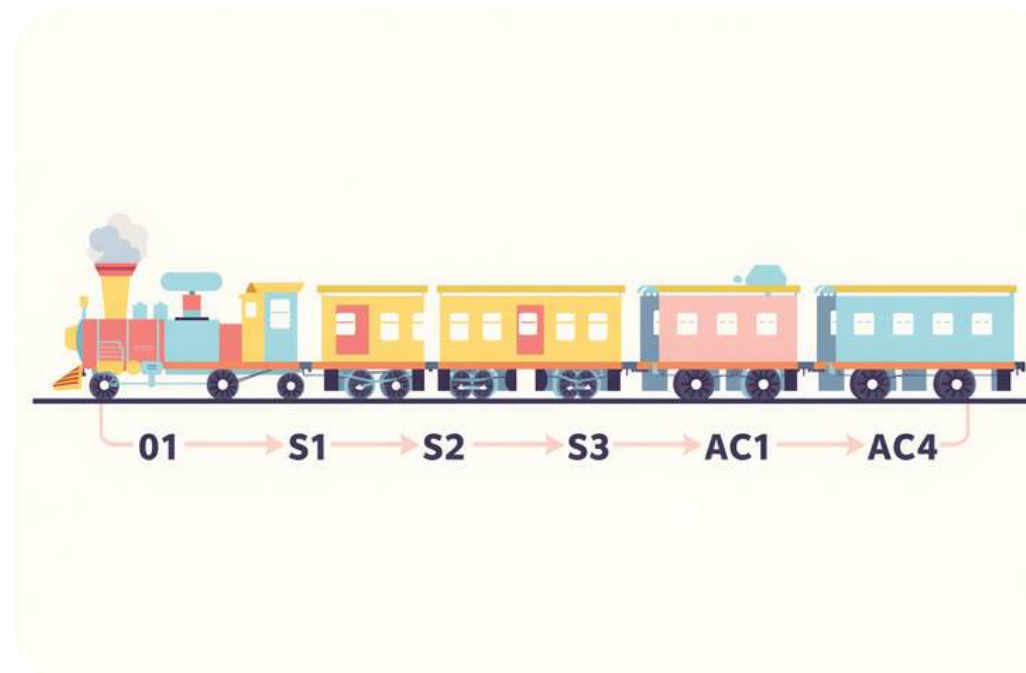




ACCESSING TUPLE ITEMS

Use index numbers to get items from a tuple!

```
print(train_coaches[0]) #  
S1  
print(train_coaches[3]) #  
AC1  
print(len(train_coaches))  
# 5  
Remember: Index starts at 0, not 1!
```





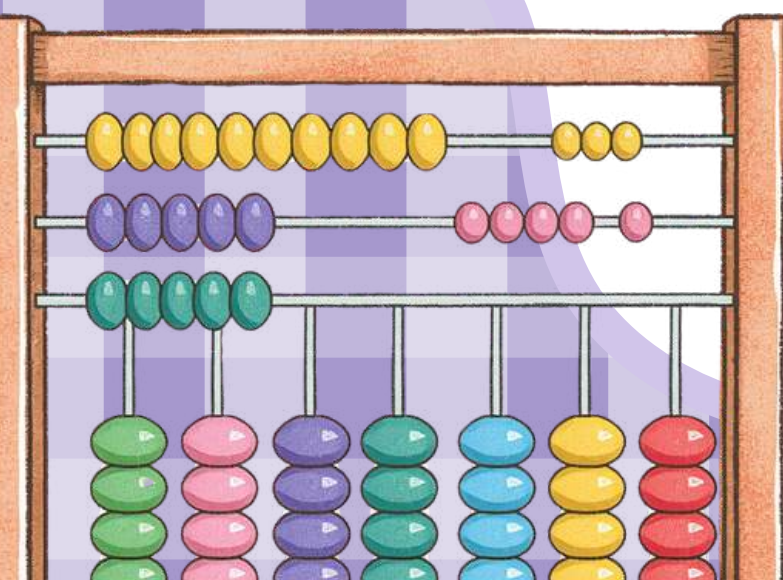
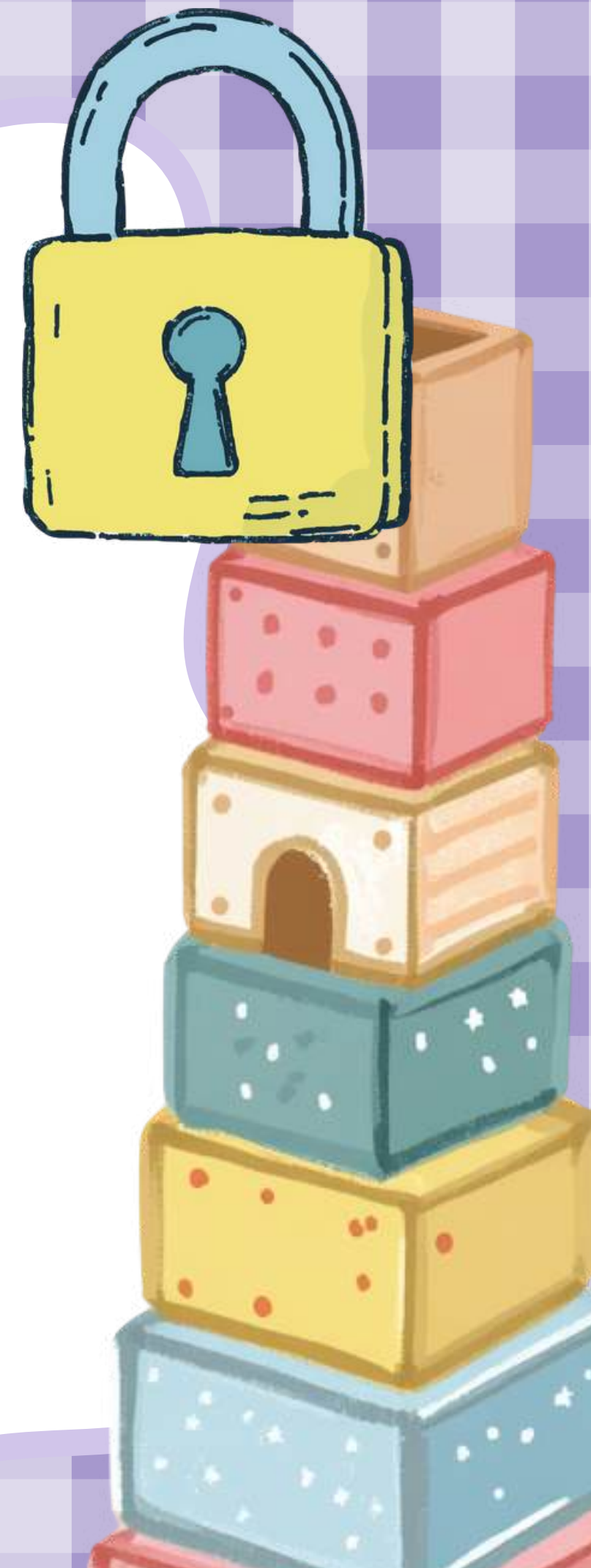
TUPLE PROPERTIES

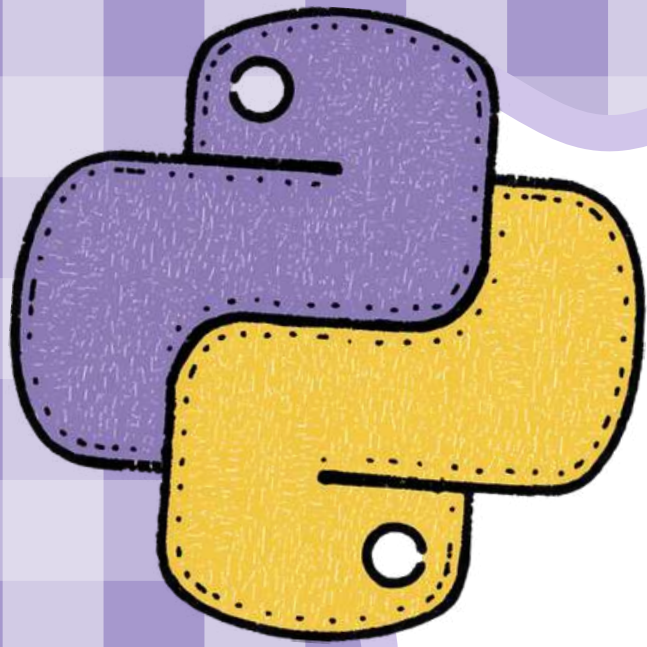
Ordered

Immutable

Duplicates OK

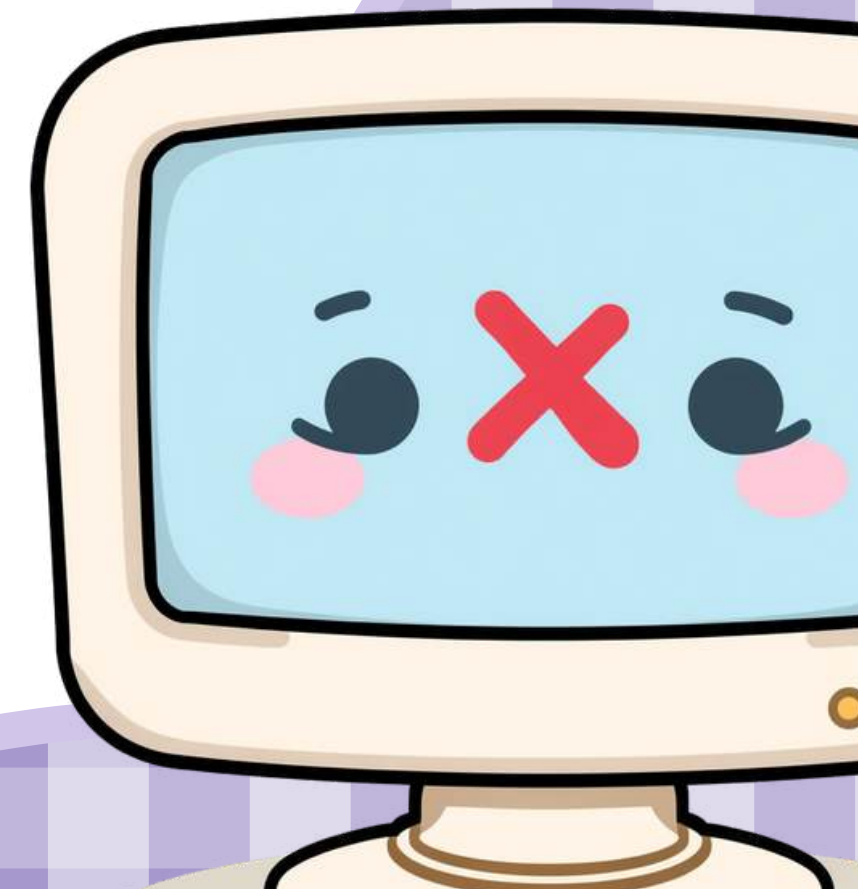
Mixed Types

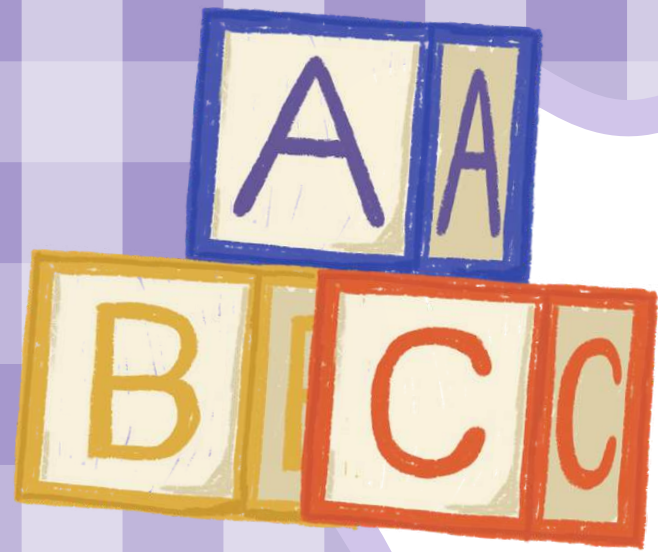




WHAT HAPPENS IF YOU CHANGE A TUPLE?

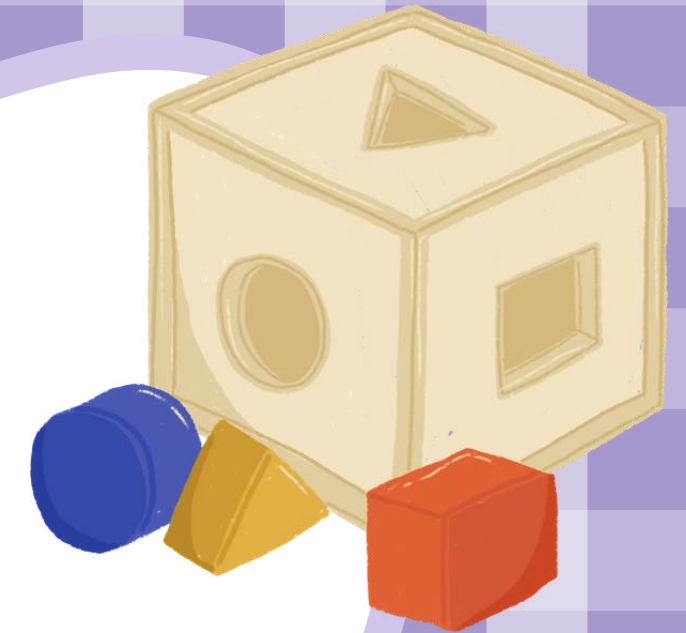
`train_coaches[0] = "B1"`
TypeError! You cannot change a tuple once it is made. Tuples are immutable, which means their items stay fixed forever!





WHAT IS A SET?

A set is a collection of items that has no order and no duplicates. Think of it like a dabba (tiffin box) with unique dals — you can have rajma, chana, and moong, but no dal appears twice!





SET ANALOGY — THE DABBA (TIFFIN BOX)

Imagine your school dabba has rajma, chana, moong, and you accidentally add rajma again. In a set, the duplicate rajma disappears automatically! Sets only keep unique items.

Another example: When your teacher takes roll call, each student's name appears only once in the register — no repeats allowed! That's exactly how sets work in Python.



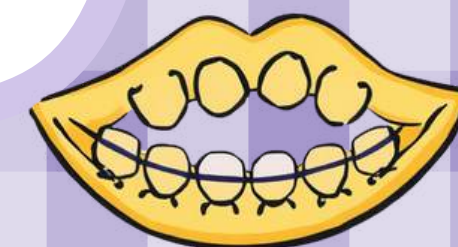
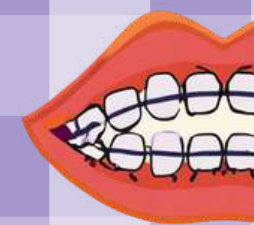
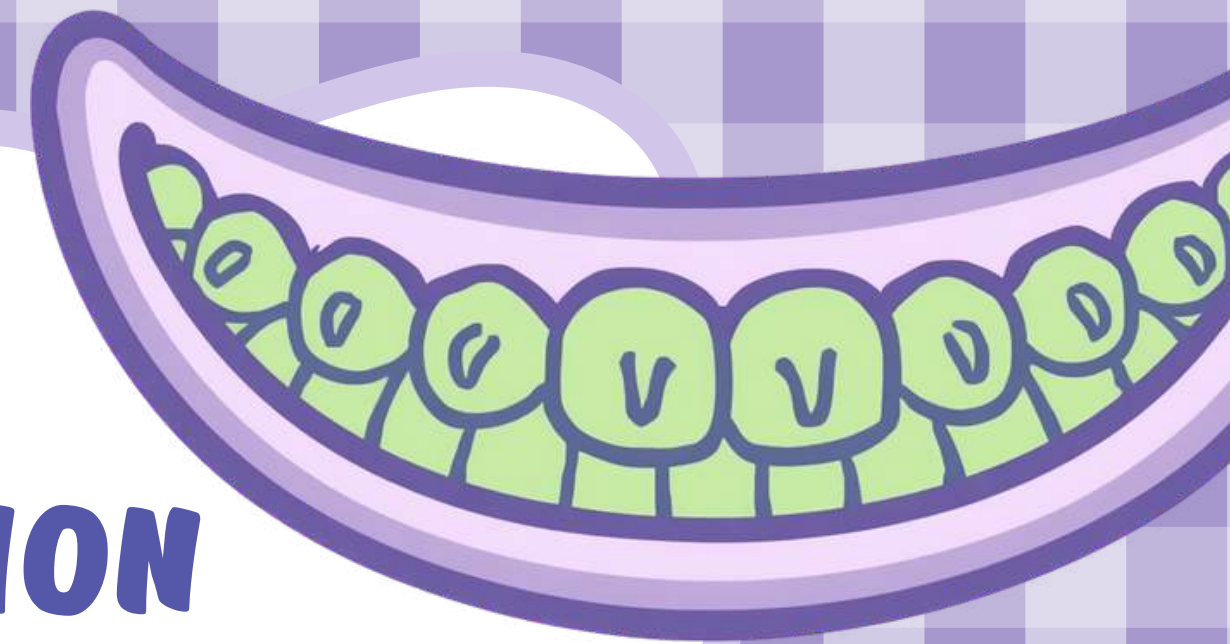


CREATING SETS IN PYTHON

Use curly braces { } to create a set:
dals = {"rajma", "chana",
"moong", "rajma"}
print(dals)
Output: {'rajma', 'chana',
'moong'}

- Use curly braces { }
- Items separated by commas
- Duplicates removed automatically!

```
python.(:>; =)  
python::set); = 1},  
nlthon.(:; set),  
{(= > shilc  
pythona = in set = set)
```



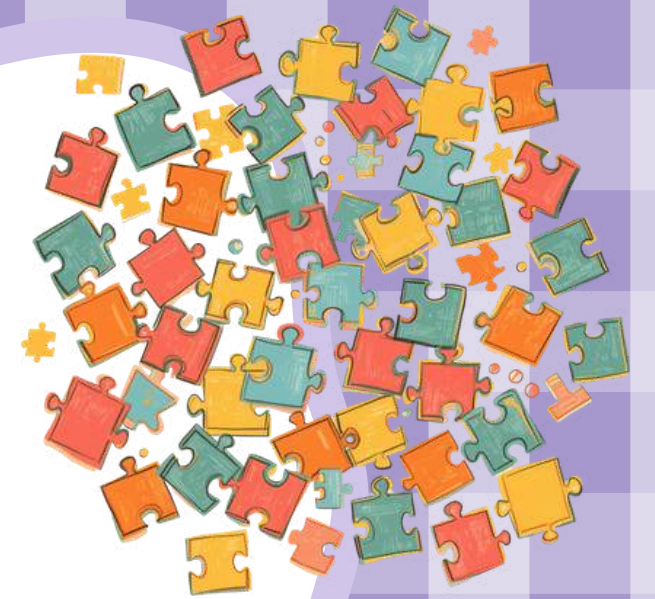
SET PROPERTIES

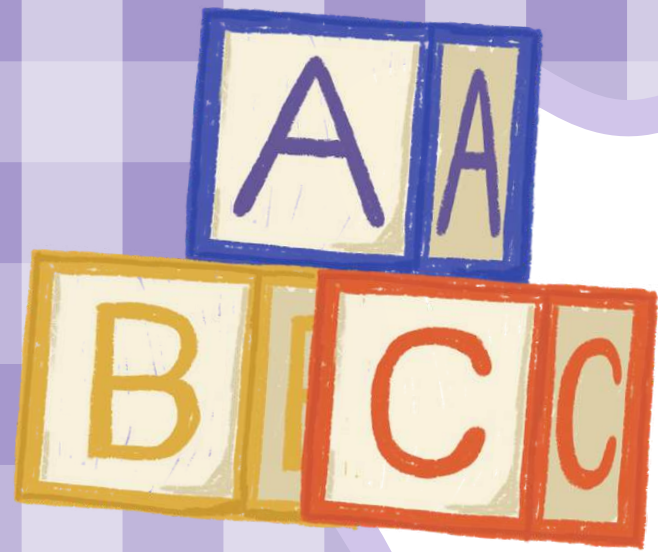
Unordered

Mutable

No Duplicates

No Indexing

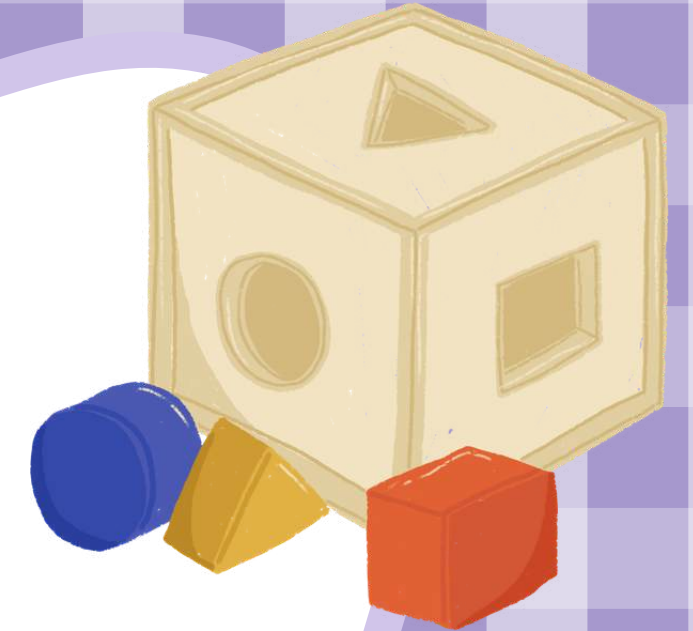


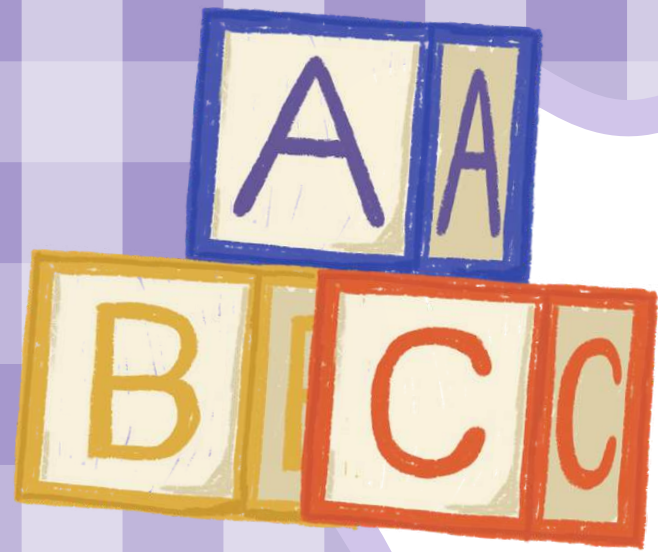


ADDING AND REMOVING ITEMS IN SETS

```
student_names = {"Priya",  
"Arjun", "Sneha"}  
student_names.add("Kiran")  
student_names.discard("Sneha")  
print(student_names)
```

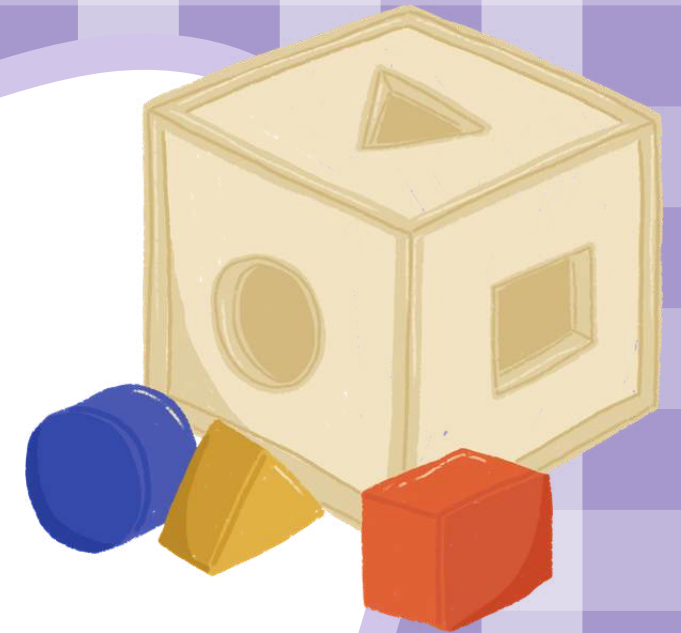
Sets can change after creation!
Use `.add()` to include new items
and `.discard()` to remove them.

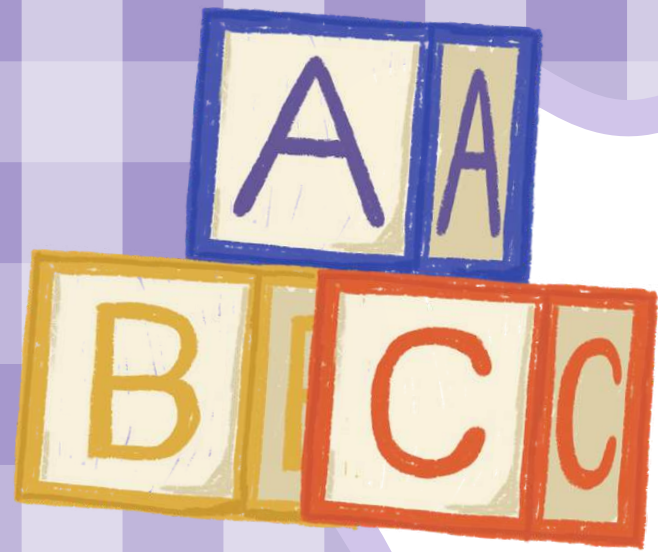




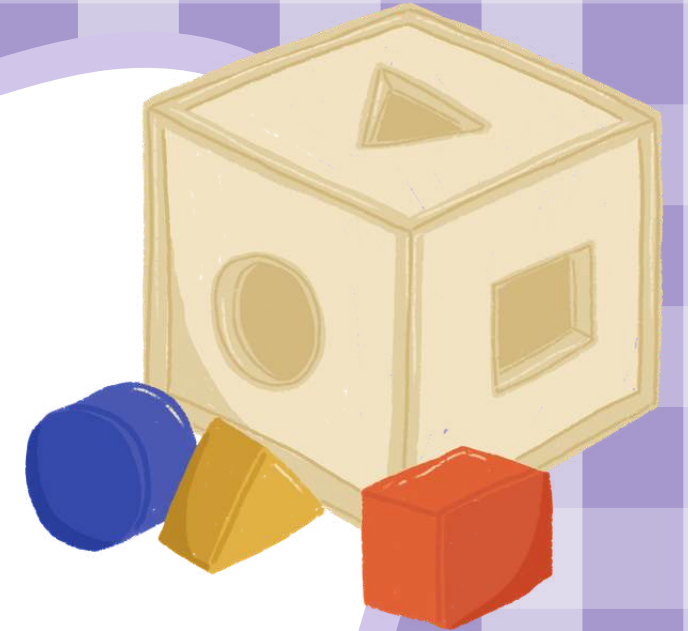
WHAT HAPPENS IF YOU TRY TO INDEX A SET?

```
student_names[0] #  
TypeError!  
Sets do not have positions,  
so you can't pick by  
number. Unlike tuples and  
lists, sets are unordered  
collections!
```





TUPLE VS SET VS LIST



Which Is Which?
Feature | List | Tuple | Set
Symbol: [] | () | { }
Ordered?: ✓ | ✓ | ✗
Can Change?: ✓ | ✗ | ✓
Duplicates?: ✓ | ✓ | ✗
Indexing?: ✓ | ✓ | ✗

flaxor xollections

List



equarsenx:



Tuple



tuple:



Set



rayniraces





WHEN TO USE TUPLES?

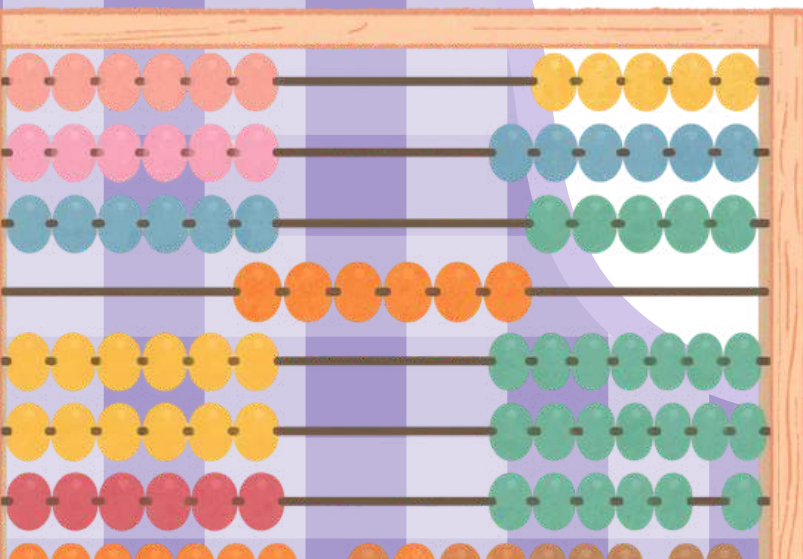
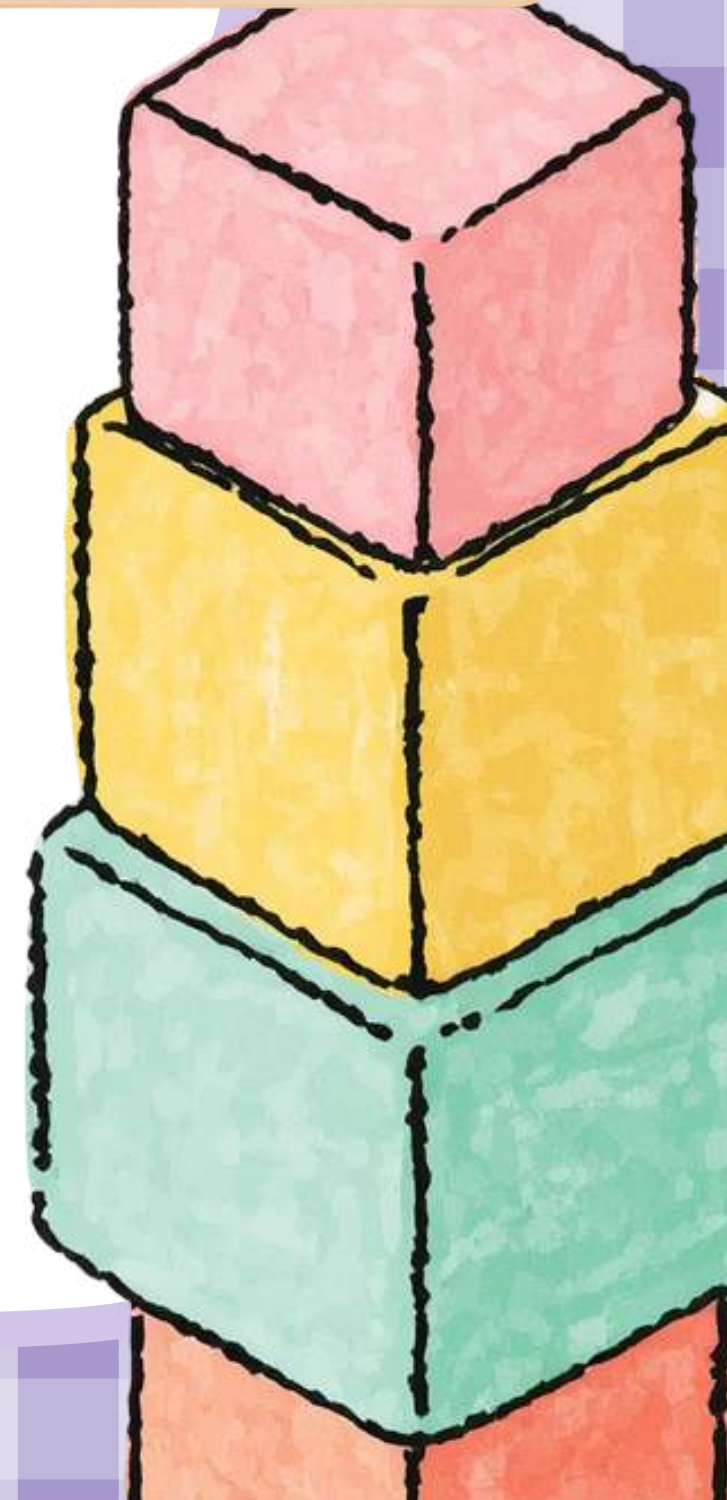


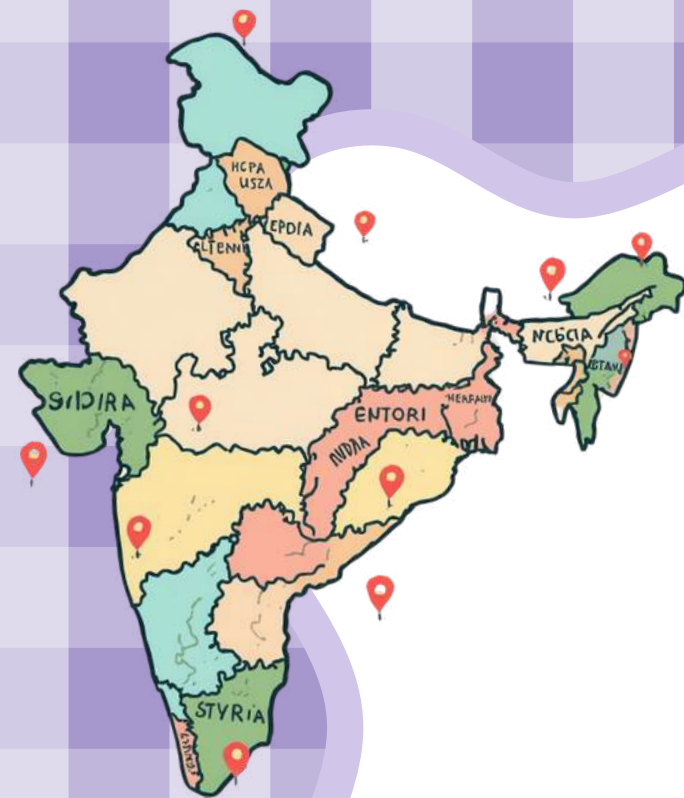
Days of the week, your date of birth

Fixed IPL teams for the season

Coordinates of your school

Indian states in order





WHEN TO USE SETS?

Unique States Visited

Languages in Class

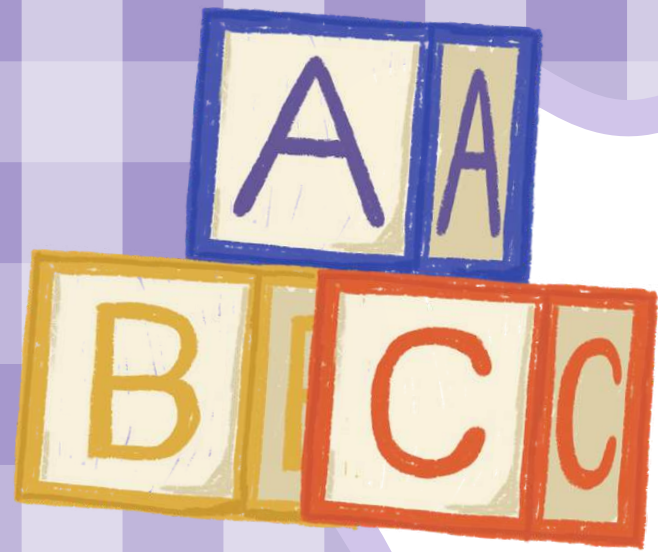
Canteen Items

Sports at School



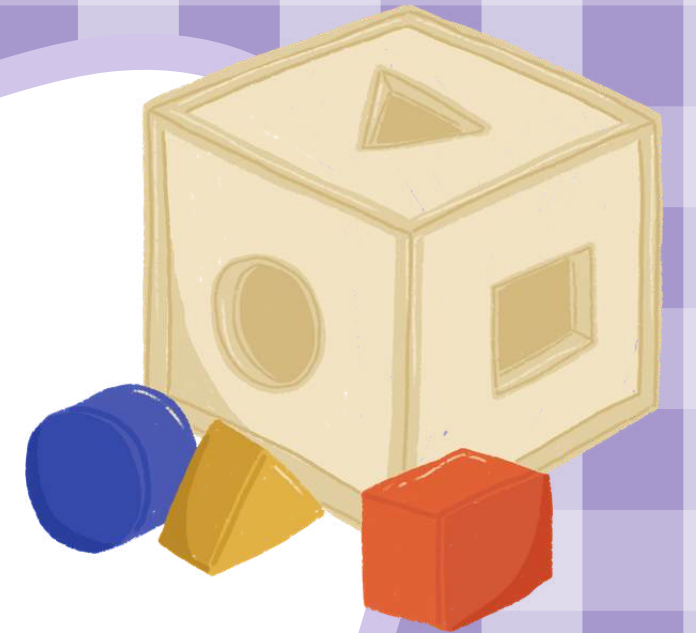
Indian
Holkay 2021

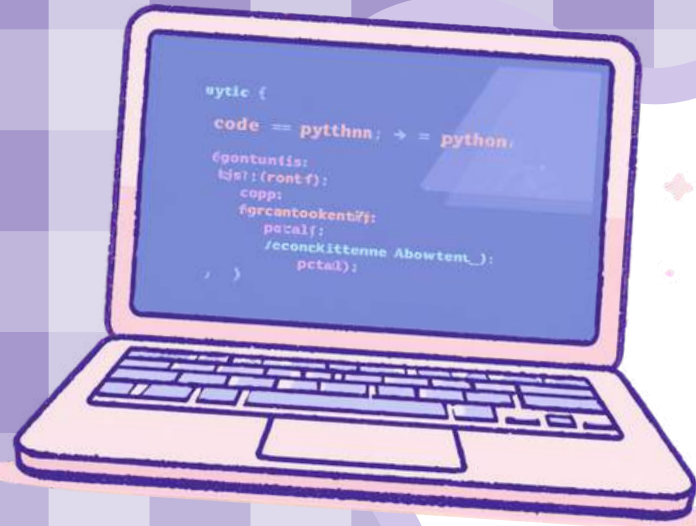




CODING ACTIVITY — MY INDIAN DATA BOX

Time to code! Create your own tuple with Indian states and a set with your favorite Indian foods. Try the fun challenge to see what happens when you run your code!





ACTIVITY STEPS 1 & 2

Step 1 — Create Your Tuple:

```
my_states =  
("Maharashtra",  
"Kerala", "Rajasthan",  
"Punjab", "Goa")  
print(my_states)  
print("First state:",  
my_states[0])
```

Step 2 — Try to Change It:

```
my_states[0] =  
"Gujarat"
```

TypeError! You cannot change a tuple once it is made.



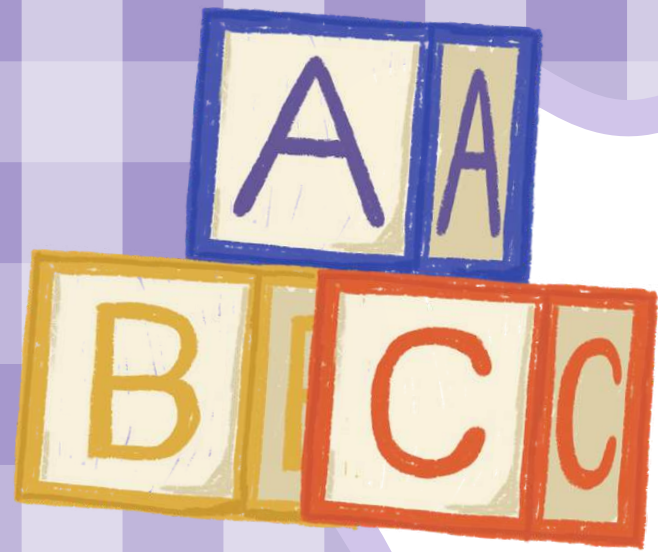


ACTIVITY STEP 3 & 4

Create Your Set:
`my_foods = {"idli",
"dosa", "paratha",
"biryani", "idli"}
print(my_foods)
print("Unique foods:",
len(my_foods))
Notice: "idli" appears
only once!`

Add & Remove Items:
`my_foods.add("pani
puri")
print(my_foods)
my_foods.discard("dosa")
print(my_foods)
Sets can change after
creation!`

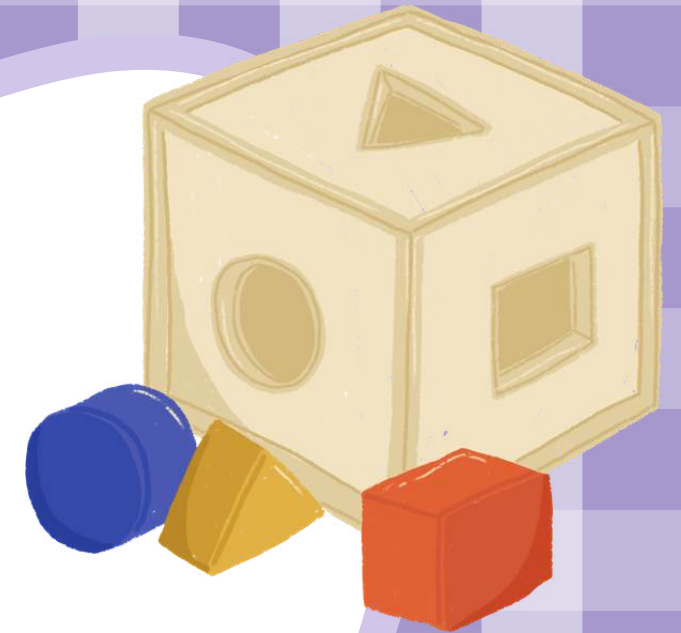




ACTIVITY STEP 5 — CHECK MEMBERSHIP

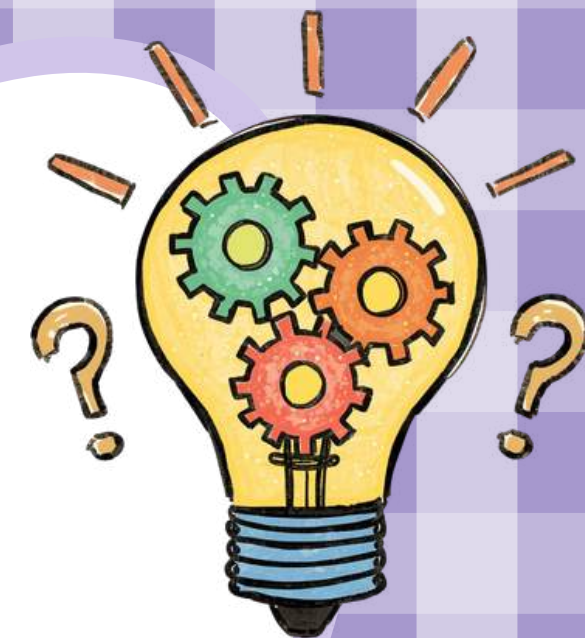
Check if items exist in your set:
`print("biryani" in my_foods)`
`True`
`print("pizza" in my_foods)`
`False`
Use 'in' to check if an item is in your set!

- ☒ True
- ☐ Biryani
- ☒ false
- ☒ asties
- ☐ fizza





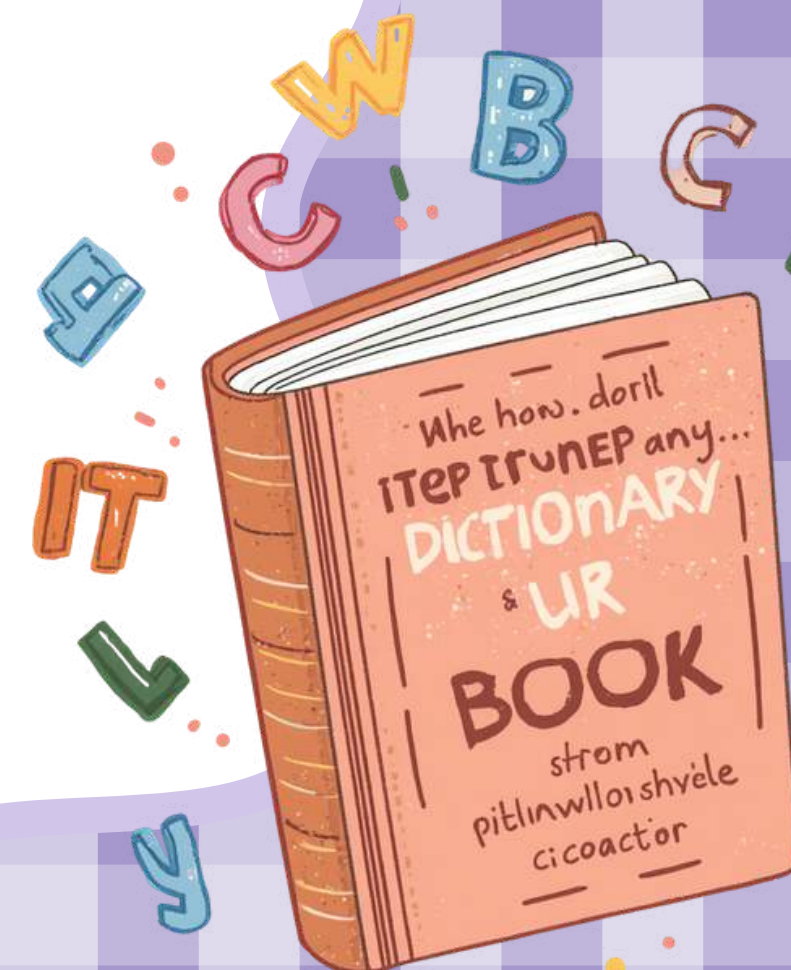
KEY VOCABULARY

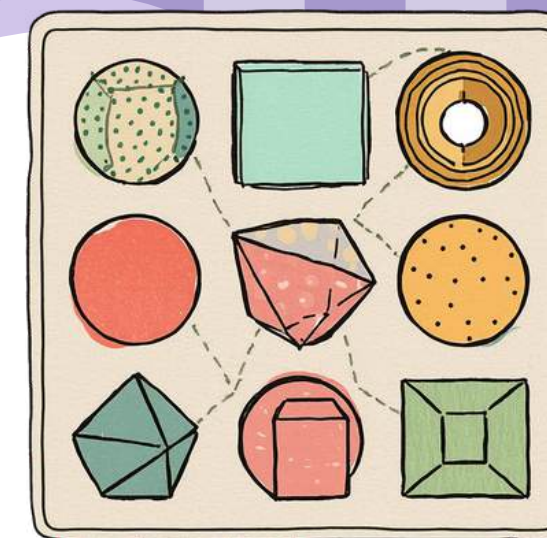


Tuple: A collection of items in a fixed order that cannot be changed, like train coaches. Set: A collection of unique items with no fixed order, like different dals in a dabba. Immutable: Something that cannot be changed once created.

Ordered: Items have a specific position (first, second, third). Unordered: Items have no fixed position, they can be in any arrangement. Index: The position number of an item in a collection, starting from 0.

Duplicate: The same item appearing more than once. In tuples, duplicates are allowed. In sets, duplicates are automatically removed to keep only unique items!





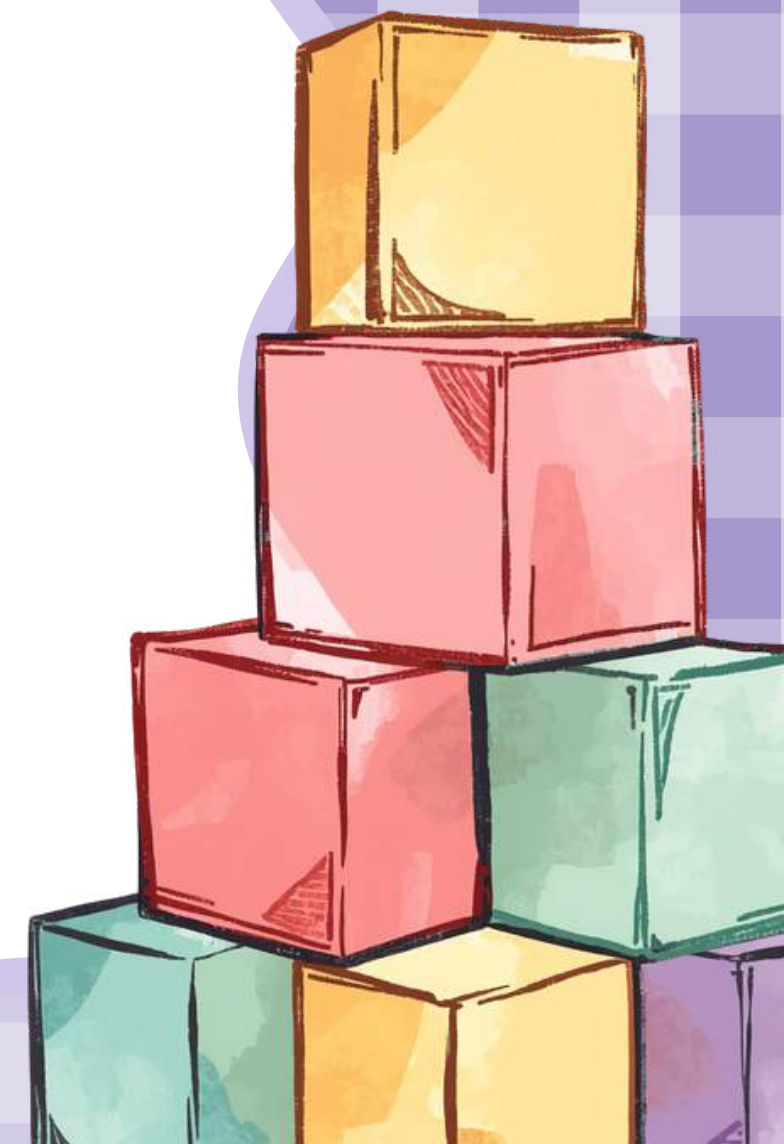
SUMMARY — WHAT HAVE WE LEARNED?

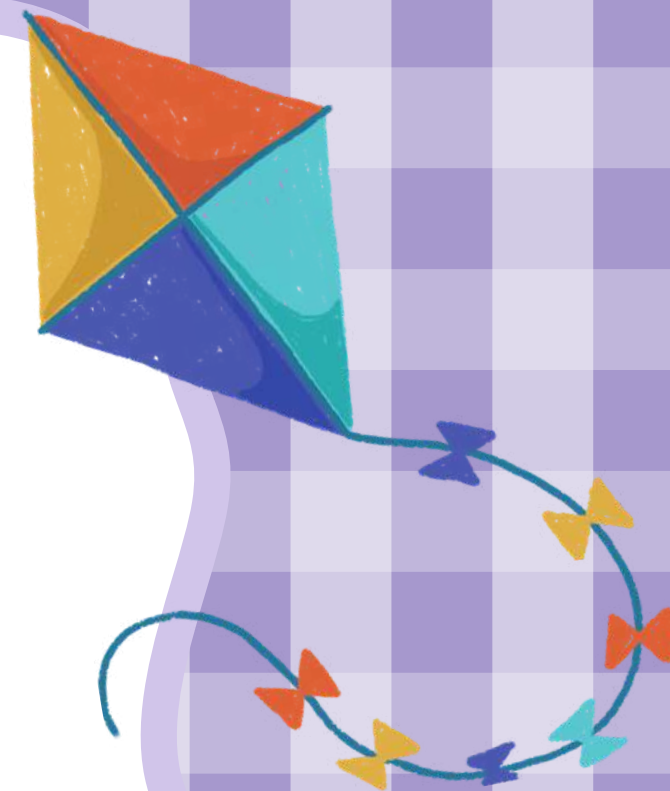
Tuples: () ordered, unchangeable, duplicates allowed — like a train with fixed coaches!

Sets: { } unordered, changeable, no duplicates — like a dabba with unique dals!

Use tuple when data should stay fixed (birth date, IPL teams)

Use set when you want unique items only (states visited, languages spoken)





THANK YOU! QUESTIONS?

